

# SAATHVIK GUBBALA

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## Education

### Carnegie Mellon University

Expected December 2028

*B.S. in Materials Science & Electrical Engineering | 3.6/4.0 GPA*

*Pittsburgh, PA*

**Relevant Coursework:** Signals & Systems, Thermodynamics of Materials, Structure of Materials, Intro to Electrical & Computer Engineering, Concepts of Mathematics, Linear Algebra, Vector Calculus, Physics 1, Fundamentals of Programming

## Experience

### Software Engineering Intern

May 2026 – August 2026

*Commvault*

*Tinton Falls, New Jersey*

- Architected a full-stack PII redaction pipeline with a FastAPI backend, combining GLiNER-based entity detection, regex, and rule-based matching to identify sensitive data across 15+ file formats (PDF, DOCX, CSV, images, etc.).
- Built a configurable redaction engine supporting semantic and full redaction modes, native-format and PDF export, and user-defined sensitive fields, exposed via both a GUI and API for production-scale document processing.
- Delivered real-time progress tracking, advanced reporting, and post-processing validation/QA checks, enabling secure, auditable redaction integrated into Commvault's Compliance Search workflow for enterprise data governance.

### Teaching Assistant – 18-100 Intro to Electrical & Computer Engineering

August 2026 – Present

*Carnegie Mellon University*

*Pittsburgh, Pennsylvania*

- Guided 180 students hands-on through breadboarding, embedded programming, and waveform analysis, troubleshooting circuits using multimeters, oscilloscopes, AD3/WaveForms, Arduino, and ESP32 to build hardware debugging skills.
- Ran weekly small-group sessions with 12 students covering circuit analysis, digital logic, MOSFETs, op-amps, I<sup>2</sup>C, and embedded systems in depth, and conducted lab checkoffs assessing conceptual understanding and circuit presentation.

### Software Engineering and Machine Learning Intern

December 2024 – June 2025

*Pfizer Inc.*

*Manhattan, New York*

- Diagnosed and resolved firewall issues through Splunk log analysis eliminating recurrent errors improving stability.
- Implemented role-based access control in Python using vulnerability scanning tools to reduce excess privilege risks.
- Developed Python automation and SQL tooling to generate database metadata and enable agents to inspect databases.

## Research and Projects

### STM32 Charger Display Interface

September 2026 – Present

- Developing an STM32-based charger display to replace Carnegie Mellon Racing's PCAN laptop workflow, enabling direct communication with the Charger Control Module over USART to read cell voltages and charging states removing delay.
- Designing the supporting PCB interface and embedded firmware for display control and CCM communication, integrating the screen within the charger's high-voltage system to simplify charging and reduce troubleshooting time overall.

### STM32 Embedded Control Board

August 2026

- Designed a custom STM32F4-based PCB in Altium used to support vehicle controls that steps down 24V battery power to 5V and 3.3V, with clock, reset, decoupling, and protection circuitry for CMU Racing's Formula SAE vehicle.
- Integrated CAN bus, USB, JTAG/SWD debugging, and SD-card logging with signal conditioning and fault protection to support ongoing firmware development, vehicle-wide communication, and onboard system diagnostics.

## Extracurricular

### FSAE Electric Energetics Team - Carnegie Mellon Racing

September 2025 – Present

*Tractive Systems, Grounded Low Voltage, Data Acquisition*

*Carnegie Mellon University*

- Member of Carnegie Mellon Racing's Energetics Tractive Systems, Data Acquisition, and Grounded Low Voltage sub-teams, supporting high- and low-voltage PCB design, sensor integration, and firmware for FSAE electric vehicle 27x.
- Contributed to PCB design, firmware, and wiring across the charger, motor, sensor, and control systems as needed, gaining hands-on hardware experience with embedded systems and vehicle electronics through cross-team support.

## Technical Skills and Interests

**Software/Languages:** Python, C++, Java, Git, Altium, MATLAB, Fusion360, Solidworks, STM32, ESP32, I2C

**Fields:** Semiconductors, Embedded Systems, PCB Design, Power Electronics, Digital Logic, Firmware Development

**Interests:** PC Building, Volleyball, Basketball, Music, Anime, Travel, Hiking, Cooking/Baking